

Knesset Protocol Journal Project

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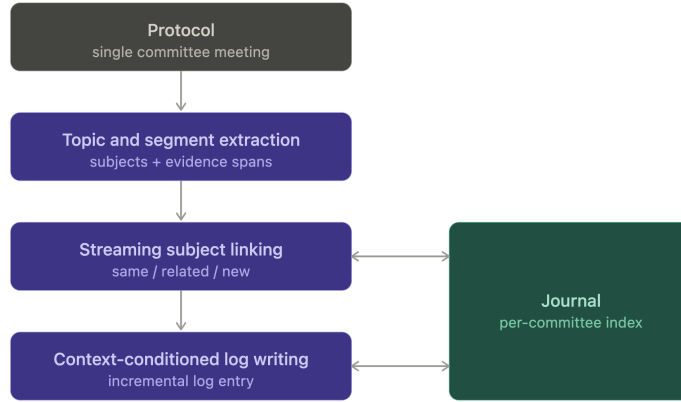


Figure 1: End-to-end pipeline

Abstract

This work studies the construction and maintenance of a longitudinal, subject-coherent journal from a stream of deliberative protocols, instantiated on Israeli Knesset committee transcripts (Figure 1). The goal is a per-committee index in which each entry corresponds to a single underlying matter and accumulates a chronological log of progress as the matter recurs across meetings. The problem decomposes into three coupled NLP tasks: topic and segment extraction, streaming subject linking, and context-conditioned log writing.

Topic and segment extraction partitions each protocol into subject-specific spans and labels each span with the matter under discussion, requiring the recovery of coherent subject boundaries in long, multi-speaker discourse and the identification of subjects at a granularity that supports tracking the same matter across meetings.

Streaming subject linking is a variant of cross-document event coreference operating online over a growing index. The setting is linguistically adversarial: ongoing matters drift lexically across meetings while distinct

matters in the same domain share most surface tokens, leaving the disambiguating signal in named entities, temporal expressions, and domain-specific identifiers rather than topical content. Anti-duplication is the central design constraint; we evaluate under the asymmetric assumption that false-merges damage the journal more than false-splits.

Context-conditioned log writing is framed as update summarization with retrieval augmentation. The journal entry for the matched subject is supplied to the model as retrieved context alongside the discussion of that subject in the current meeting, and the generation is constrained to produce only the incremental contribution: new developments, decisions, status changes, and cross-references to prior entries. The retrieved log does more than condition the output topically; it records what has already been written so that the matter is not re-summarized from inception. Faithfulness is therefore defined relative to a chain of prior summaries rather than to a single source document, distinguishing this setting from standard query-conditioned RAG.

We evaluate each task on a manually annotated subset of committee meetings, and the resulting journal end-to-end through human assessment of longitudinal coherence and faithfulness.

First Milestones

1. **Annotated evaluation set.** Manually label a small held-out subset with gold subjects per meeting and gold cross-meeting linking decisions (which new subject corresponds to which prior journal entry, if any). Without this, no quantitative comparison is possible.
2. **Topic and segment extraction.** Implement a first pass that partitions each protocol into per-subject spans and labels each span with a canonical subject title plus key entities. Choose the simplest approach consistent with what the dataset inventory shows.
3. **Streaming subject linking.** Implement and compare two methods for resolving each newly extracted subject against the existing journal.
 - (a) **Similarity threshold baseline.** Embed each subject (title + summary + entities) with a Hebrew-capable multilingual encoder, take the top-1 nearest neighbor from the journal, and treat it as the same matter if cosine similarity exceeds a tuned threshold. Pure dense retrieval with no LLM, establishing the floor.
 - (b) **Retrieve-then-verify.** Use top- K dense retrieval to shortlist candidates and an LLM verifier to classify each new subject as same, related, or new given the candidate entry’s full prior timeline. The proposed approach, isolating the value of the verifier over pure similarity.
4. **Asymmetric evaluation harness.** Compute precision, recall, and F1 with separate accounting for false-merge and false-split errors. Run both linking methods on the evaluation set and report the gap.